GENX320 Event-Based Vision Sensor

Research Progress Presentation

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Project objective and research motivation

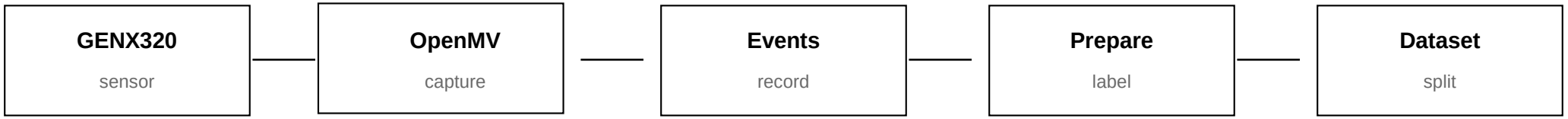
The project moves from understanding the sensor to collecting a usable event-based dataset.

- The GENX320 is an event-based vision sensor that reports changes in brightness instead of capturing only complete images.
- Event data can preserve fine timing information and reduce repeated data from parts of a scene that do not change.
- The project investigates how to access GENX320 data through the OpenMV Cam H7 Plus and OpenMV MicroPython.
- The practical goal is to define a repeatable recording process and collect labeled sequences for a future computer-vision task.

Overall system: sensor-to-dataset workflow

The workflow below is planned; current verified work is the initial board connection and grayscale snapshot test.

- 1. Connect the GENX320 module to the OpenMV Cam H7 Plus and initialize it with the OpenMV MicroPython API.
- 2. Select histogram mode for image-like output or raw event mode to preserve individual event timestamps and coordinates.
- 3. Record the selected output to the SD card or stream it to the PC, while saving sensor settings and scene conditions.
- 4. Visualize, label, and convert the recordings into a consistent format for later model development.
- 5. Split by recording or trial into training, validation, and test sets to prevent data leakage.



GENX320: published technical specifications

These values come from the investigation report and manufacturer / SDK documentation, not from my own measurements.

PROPERTY	PUBLISHED VALUE
Resolution	320 × 320 pixels
Pixel size	6.3 × 6.3 µm
Timestamp precision	1 µs, row-level timestamp precision in sensor documentation
Pixel latency	Less than 150 µs at 1,000 lux
Dynamic range	Above 140 dB on product page; above 120 dB on SDK page
Power	36 µW ultra-low-power mode; approximately 3 mW typical
Embedded filters	AFK (anti-flicker), ERC (event-rate control), STC (spatio-temporal contrast)

Note: timestamp precision and pixel latency are different properties. Dynamic-range values need datasheet-revision confirmation.

Two GENX320 output modes in OpenMV

The output mode determines how much event-level information is retained.

HISTOGRAM MODE



Events are accumulated into a 320 × 320 grayscale image.
Documented output rate: approximately 20–350 FPS.

RAW EVENT-STREAM MODE

type	sec	msec	usec	x	y
ON	0	12	340	84	106
OFF	0	12	515	85	106
ON	0	12	702	86	106

Individual events are returned in a uint16 array with six columns: type, sec, msec, usec, x, and y.

Histogram mode is easier to inspect. Raw event mode preserves individual timestamps, positions, and polarity for event-based processing.

Research progress | Completed: November 1–8

Task 1 was a desk-based technical investigation, not a hardware experiment.

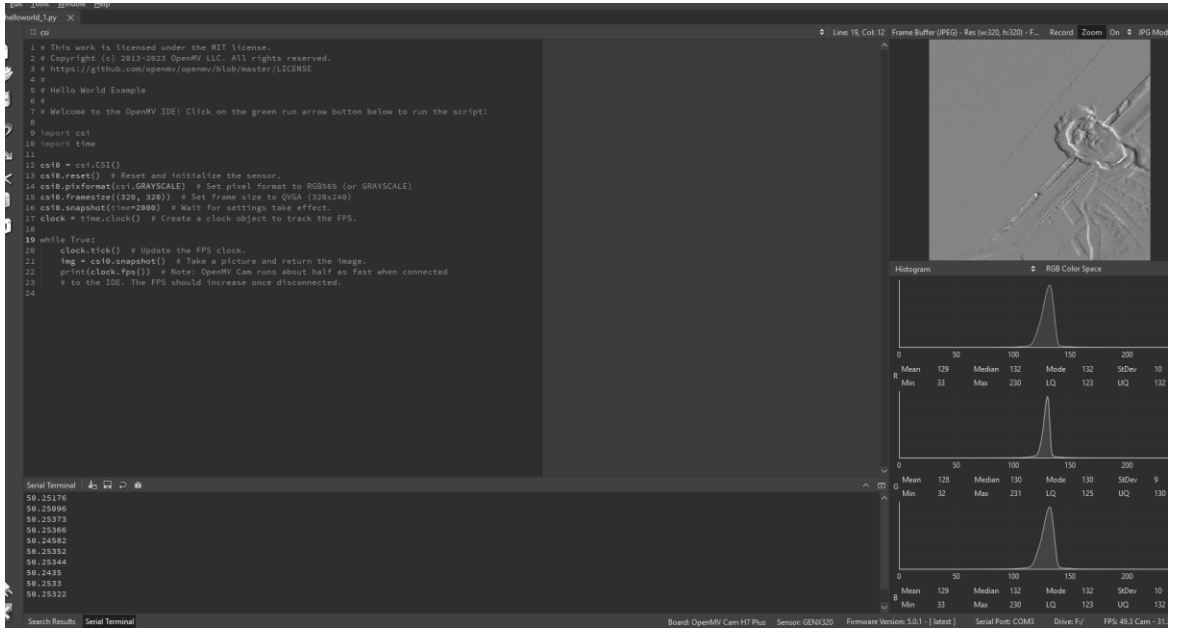
- I investigated how the GENX320 detects local changes in logarithmic brightness and produces ON and OFF events.
- I summarized the published sensor specifications, including resolution, pixel size, timestamp precision, latency, power, and embedded filters.
- I compared histogram mode with raw event-stream mode and reviewed the OpenMV GENX320 and camera-sensor APIs.
- I reviewed the differences between event-based and frame-based vision, including timing, motion blur, and data generation.
- I reviewed the Event-Based Vision survey and existing public datasets, and confirmed that the reviewed datasets are not GENX320 recordings.

Status: Task 1 investigation completed. The full datasheet revision check and report commit remain checklist items in the report.

Research Progress | Completed

November 1–8 investigation, plus verified basic board setup.

- Completed the GENX320 sensor investigation.
- Reviewed the OpenMV API and both output modes.
- Connected the OpenMV Cam H7 Plus in OpenMV IDE.
- Ran a grayscale snapshot test at about 49.3 FPS.



OpenMV IDE screenshot: sensor view, histogram, and serial output

Setup shown: firmware 5.0.1 | COM3 | F:/ drive | ~49.3 FPS snapshot test

Research progress | Current work: November 9–15

The main objective this week is to verify the GENX320-specific event functions.

- I am setting up the OpenMV Cam H7 Plus, OpenMV IDE, and compatible firmware for the GENX320 module.
- I will run the official MicroPython examples and confirm that histogram output responds to controlled motion.
- I will test raw event mode and inspect event type, timestamps, coordinates, and event-buffer contents.
- I will visualize static scenes, slow motion, fast motion, and changes in illumination.
- I will test bias and ROI settings, record short sample sequences, and define the target task, classes, and recording protocol.

Future work | Dataset collection: November 16–24

After the output modes and recording procedure are verified, I will collect labeled sequences.

- I will write MicroPython scripts on the OpenMV Cam to record the selected GENX320 output mode.
- I will collect and label event histograms, event-derived frames, or raw event streams according to the final task.
- I will save recordings to the SD card or stream them to the PC, depending on the verified transfer method.
- For every recording, I will log the label, recording conditions, sensor settings, and relevant environment information.
- I will convert the collected data into a training-ready format and create training, validation, and test splits without data leakage.

Schedule

November 2026

Task	Nov 1–8	Nov 9–15	Nov 16–24	Nov 25–30
GENX320 investigation				
OpenMV setup and tests				
Dataset collection				
Final report and scripts				

- Task 1: Sensor investigation completed.
- Task 2: OpenMV setup and event-mode tests are in progress.
- Task 3–4: Dataset collection and final report are planned.

Thank you for your attention!